

FEASIBILITY STUDY

Courier Services Management and Tracking Platform

[Working title — no branded product name has been assigned yet]

Executive Summary

The organisation's package collection point currently manages a growing volume of incoming courier packages using Microsoft SharePoint together with a largely manual administrative process. Packages are logged by hand, located through manual search when a customer arrives, and income is reconciled by manually working through receipts. As parcel volume grows, this creates increasing delays, a higher chance of reporting error, and no proactive way to inform customers that their package is ready.

This feasibility study evaluates two ways to address this: continuing with the current SharePoint-based process, or developing a Courier Services Management and Tracking Platform ("the Platform"), a purpose-built web application covering package intake (via barcode scanning), status tracking, automatic customer notification, a staff dashboard, and automatic income reporting based on a flat collection fee per package.

Each option is assessed against technical, financial, operational, and human-resource feasibility, with an estimated first-year cost breakdown. The Platform is found to be the more feasible long-term option: it directly resolves the operational bottlenecks in the current process, and its estimated costs are achievable within the scope of this project. While it requires more upfront effort than doing nothing, careful phasing and a clearly scoped MVP can manage the associated risks around time, resourcing, and implementation complexity.

Problem Statement

Business Problem

The organisation currently manages package intake, storage, and collection using SharePoint and manual administrative processes. Packages are logged by hand, located by manual search when a customer arrives, and income is reconciled by manually working through receipts. As parcel volume grows, this creates increasing delays, a higher chance of reporting error, and no proactive way to tell customers their package is ready.

- Inefficient logging – capturing each package manually in SharePoint takes staff time proportional to volume, with no way to speed this up during busy periods.
- Slow retrieval – locating a specific package on collection requires manual search through SharePoint and, in some cases, physical storage.
- Manual, error-prone reconciliation – total income is calculated by manually totalling receipts rather than through the system.
- No proactive notification – customers are not told automatically when their package is ready, causing wasted trips or prolonged uncollected storage.
- No throughput visibility – staff have no live view of how many packages are outstanding versus already collected.

NOTE FOR LECTURER/STUDENTS: *As with the business case, please confirm current parcel volumes and roughly how long the main manual tasks take today — this is the single most useful figure for making the cost-benefit comparison in this study realistic rather than illustrative.*

Business Opportunities

- Barcode-based logging and lookup can materially cut the staff time spent per package.
- Automatic SMS/WhatsApp notifications remove the need for staff to individually contact customers.
- Automatic income reporting removes manual reconciliation and the errors that come with it.
- A live collected-vs-uncollected dashboard gives staff and management visibility that does not exist today.
- A well-structured MVP creates a foundation that can be extended later without a full redesign.

Requirements Statement

Business Drivers

- Reduce administrative time spent on package logging, retrieval, and reporting.
- Provide real-time visibility of packages collected versus uncollected.

- Automate the flat-rate income calculation so totals are available on demand.
- Notify customers automatically once a package is ready for collection.

Business Requirements

Business Problem / Opportunity	Project System Requirement
Manual parcel logging is slow	Support fast intake via barcode/QR scan (dedicated scanner or webcam).
Staff cannot quickly find a parcel	Support lookup by tracking number, customer name, or phone number.
Income reconciliation is manual	Automatically calculate income (flat rate × parcels collected) for any date range.
No proactive customer notification	Automatically send SMS/WhatsApp when a parcel is marked ready.
No visibility of throughput	Provide a dashboard of parcels collected vs uncollected.
Personal data must be handled responsibly	Comply with POPIA — restricted access and a defined retention period for customer contact details.

Feasibility Assessment

This section compares building the Platform against continuing with the current system, assessed against technical, financial, operational, and human-resource feasibility. Each option is examined on its own merits before being ranked against the other in the sections that follow.

Solution 1: The Courier Services Management and Tracking Platform

Description

The Platform is a web application covering package intake (via barcode scan or manual entry), status tracking, automatic customer notification, a staff dashboard, and automatic income reporting based on a flat collection fee per package. It is designed to be built and delivered in a staged manner, with the core intake-lookup-report workflow prioritised ahead of the notification and dashboard modules.

Drawbacks

- Requires development and testing time before it can fully replace the current process.
- Introduces an ongoing cost for SMS/WhatsApp messaging and hosting, which the current system does not have.
- Barcode scanning reliability (whether via dedicated scanner or webcam) needs to be proven in testing before go-live.

Benefits

- Cuts staff time per package through barcode-based logging and fast search, instead of manual entry and manual search.
- Removes manual income reconciliation entirely.
- Reduces wasted customer trips through automatic ready-for-collection notifications.
- Gives staff a live operational view that supports planning around storage and peak intake periods.

Feasibility Scores

Criterion	Score (1-5)	Explanation
User Friendly	4	Scan-based intake is simpler than manual SharePoint entry and manual search.
Budget Consciousness	4	Low development cost; main ongoing cost is hosting and messaging.
Implementation	2	Intake, notification, dashboard, and reporting modules all need to be built from scratch.
Time Consciousness	3	An MVP is realistic within a project timeline; full feature set will take longer.
Adaptability	4	Modular design allows later additions without a redesign.
End User Instruction	4	Simpler workflow than the current process should mean short training sessions.
Human Resources	3	Delivered by a small development team; requires coordination but no external hires.

Risks

Time Consciousness

Delivery timelines may be affected by team availability and competing priorities. A rushed build increases the risk of incomplete or unreliable features at launch.

Implementation

Barcode scanning and messaging-API integration are new territory for a first build and may take longer than estimated to make reliable across both scanning methods.

Messaging Cost

SMS/WhatsApp costs scale with parcel volume; an underestimated volume could push operating costs above budget once the system is live.

Estimated Cost Breakdown

Development and Technical Setup

Item	Estimated Cost	Notes
Cloud hosting (1 year)	R12 000	Placeholder — based on a mid-tier hosting plan for the web app and database.
Domain registration	R120	Annual fee for a .co.za domain.
SSL certificate	Free	Free via Cloudflare with auto-renewal.
Barcode scanner hardware (optional)	R800 – R2 500 per unit	Only required if dedicated scanners are chosen over webcam-based scanning — see Implementation Note in the business case.
Subtotal	R12 120 + optional hardware	

Notification Service (SMS/WhatsApp)

Item	Estimated Cost	Notes
Messaging API setup	R0 – R1 500	Placeholder — most gateways (e.g. Twilio, WhatsApp Business API) carry no or low setup fees.
Per-message cost	R0.30 – R0.90 per SMS (placeholder)	Actual rate depends on provider and volume — a real quote is needed to finalise this.
Estimated monthly cost	[Volume × rate]	Cannot be finalised until parcel volume is confirmed.

NOTE FOR LECTURER/STUDENTS: This section cannot be finalised without (a) a confirmed or reasonably estimated parcel volume and (b) a quote from a specific SMS/WhatsApp provider. Please provide these, or have students obtain provider quotes as part of the assignment.

Implementation and Data Migration

Item	Estimated Cost	Notes
Data migration from SharePoint	R2 000	Placeholder — exporting and importing existing open parcel records.
Backup and recovery setup	R1 500	Cloud backup configuration.
Subtotal	R3 500	

Training and Change Management

Item	Estimated Cost	Notes
Staff training sessions	R1 500	Short in-person or online sessions run by the development team.
Training materials	R500	Digital quick-reference guide.
Subtotal	R2 000	

Total (excluding messaging and optional hardware, pending confirmation)

Category	Price
Development and Technical Setup	R12 120
Implementation and Data Migration	R3 500
Training and Change Management	R2 000
Contingency (10%)	R1 762
Grand Total	R19 382 + messaging costs + optional scanner hardware

Conclusion

The Platform offers a system built around the organisation's actual workflow rather than a generic list-based tool. First-year fixed costs are modest and largely one-off, with the main variable cost being messaging, which scales with parcel volume and needs a confirmed figure to finalise fully. With reasonable planning, the Platform is achievable within scope and directly resolves the operational issues identified above.

Solution 2: Keep Current System

Description

This option continues using SharePoint lists for parcel logging, with staff manually searching for packages and manually reconciling receipts to calculate income totals.

Drawbacks

- Administrative time per package remains high and scales with volume.
- Income reporting stays manual, slow, and error-prone.
- Customers continue to receive no automatic ready-for-collection notification.
- Staff have no live view of packages outstanding.

Benefits

- No development cost or delivery risk.
- No retraining required.
- No new ongoing software or messaging cost.

Feasibility Scores

Criterion	Score (1-5)	Explanation
User Friendly	3	Staff already know it, but search and reconciliation stay slow.
Budget Consciousness	4	No new spend required in the short term.
Implementation	5	Nothing to implement — already in use.
Time Consciousness	5	No implementation time needed.
Adaptability	1	SharePoint lists are difficult to extend with automation.
End User Instruction	5	No new training required.
Human Resources	2	Manual effort continues to consume staff time on every parcel.

Risks

Operational Inefficiency

Manual processes and workarounds may seem manageable now, but they are likely to become an increasing bottleneck as parcel volumes grow, leading to slower service and greater staff and customer frustration over time.

Data Handling and Compliance

Ad-hoc manual processes make it harder to enforce consistent, POPIA-aligned handling of customer information such as names and phone numbers, compared to a system with structured access control.

Reporting Accuracy

Manual income reconciliation carries an ongoing risk of arithmetic or transcription error, a risk that grows in proportion to parcel volume rather than shrinking over time.

Estimated Cost Breakdown

Item	Estimated Cost	Notes
SharePoint licensing	R0 (placeholder)	Assumed already covered under an existing organisational Microsoft 365 agreement — please confirm.
Ongoing staff time (opportunity cost)	Not quantified in Rand terms	The real cost of this option is staff hours, not a fixed line-item spend — worth estimating in hours/week for a fair comparison.

NOTE FOR LECTURER/STUDENTS: *Please confirm whether SharePoint/Microsoft 365 is already licensed centrally (likely at no direct cost to the collection point) so students can present a fair, like-for-like cost comparison against the Platform.*

Conclusion

Keeping the current system avoids new spend and disruption in the short term, but does not resolve the operational problems driving this project. Its main real cost is staff time, which continues indefinitely and grows with parcel volume rather than reducing over time.

Feasibility Ranking

Criterion	The Courier Services Management and Tracking Platform	Keep Current System
User Friendly	4	3
Budget Consciousness	4	4
Implementation	2	5
Time Consciousness	3	5
Adaptability	4	1
End User Instruction	4	5
Human Resources	3	2
Total	24	25

Feasibility Results

Solution 1 (The Platform): technically feasible using a standard, well-documented stack (React, Node.js, MySQL); the main technical risk lies in barcode-scanning reliability and messaging-API integration, both new to a first-time build team. Financially, first-year one-off costs are modest, though the messaging cost cannot be fully finalised without a confirmed parcel volume. Operationally, it directly resolves every problem identified in the problem statement, including notification, reporting, and throughput visibility.

Solution 2 (Keep Current System): technically trivial, as nothing new needs to be built, but it does not resolve any of the identified problems. Financially it appears cheapest on paper, but its real cost — staff time spent on manual logging, retrieval, and reconciliation — is ongoing and grows with volume rather than being a fixed, one-off spend.

Recommendation

The Courier Services Management and Tracking Platform is the recommended solution. It scores lower on short-term implementation ease and time-to-value, which is expected for any newly built system, but it is the only option that resolves the operational bottlenecks driving this project: slow logging and retrieval, manual income reconciliation, absent customer notifications, and no throughput visibility. A phased MVP rollout — starting with intake, lookup, and reporting, then adding notifications and the dashboard — keeps the project realistically achievable within the available timeline.

Conclusion

The Platform provides the better long-term balance of cost, control, and operational improvement. With a confirmed parcel volume, a chosen messaging provider, and a decision on barcode scanning method, this feasibility study can be finalised with real figures rather than the placeholder estimates used throughout this document.

Appendix: Assumptions and Information Still Required

This document has been prepared as a detailed working template. The following items should still be confirmed to finalise the feasibility study with real figures:

- Current parcel volume (per day/week/month), to size messaging costs and validate time-savings claims.
- The flat collection fee amount.
- Confirmation of SharePoint/Microsoft 365 licensing cost (likely already centrally licensed).
- Preferred/approved SMS or WhatsApp provider, or a messaging budget ceiling.
- Whether barcode scanners are already owned, and whether workstations have webcams.
- Overall budget ceiling and target delivery timeline for the project.